

Fundamentals Report: AI & Machine Learning

1. Defining AI and ML

Artificial Intelligence (AI) is the broad field of building systems that can perform tasks normally requiring human intelligence — reasoning, perception, language understanding, planning, and decision-making. AI is the umbrella discipline; it includes rule-based expert systems, search algorithms, robotics, and learning-based methods.

Machine Learning (ML) is a subset of AI focused specifically on systems that improve their performance on a task by learning patterns from data, rather than following explicitly hand-coded rules. Every ML system is a form of AI, but not every AI system uses ML (e.g., a simple chess engine using minimax search is AI but not ML).

Key Distinction

Aspect	AI	ML
Scope	Broad field — any machine intelligence	Subset of AI — learning from data
Method	Can be rule-based or learned	Always data-driven
Example	A GPS route planner using fixed rules	A spam filter that learns from labeled emails

2. Narrow AI vs. General AI

Narrow AI (Weak AI): Designed to perform a single task or a limited set of related tasks extremely well — e.g., voice assistants, recommendation engines, image classifiers, language models. Almost all AI in use today, including ML systems, falls into this category. It has no genuine understanding outside its trained domain.

General AI (Strong AI / AGI): A hypothetical system with human-level (or beyond) reasoning, adaptability, and understanding across *any* intellectual task, without being specifically trained for it. AGI does not currently exist; it remains a research goal and subject of active debate regarding feasibility and timeline.

3. The Three Primary Types of Machine Learning

A. Supervised Learning

The model learns from labeled data (input-output pairs) to predict outcomes on new, unseen inputs.

Common algorithms: Linear/Logistic Regression, Decision Trees, Random Forests, Support Vector Machines (SVM), Gradient Boosting (XGBoost), Neural Networks.

Use cases:

- 1 **Email spam detection** — classifying emails as spam/not spam based on labeled examples.
- 2 **House price prediction** — regression models predicting price from features like size and location.

B. Unsupervised Learning

The model finds hidden structure or patterns in data that has no labels.

Common algorithms: K-Means Clustering, Hierarchical Clustering, Principal Component Analysis (PCA), DBSCAN, Autoencoders.

Use cases:

- 1 **Customer segmentation** — grouping customers by purchasing behavior for targeted marketing.
- 2 **Anomaly detection** — identifying unusual network traffic patterns that may indicate fraud or intrusion.

C. Reinforcement Learning (RL)

An agent learns by interacting with an environment, taking actions, and receiving rewards or penalties, aiming to maximize cumulative reward over time.

Common algorithms: Q-Learning, Deep Q-Networks (DQN), Proximal Policy Optimization (PPO), Actor-Critic methods.

Use cases:

- 1 **Game-playing agents** — systems that learn to play board games or video games at superhuman levels.
- 2 **Robotics control** — robots learning locomotion or manipulation tasks through trial and error in simulation.

Summary

AI is the goal (intelligent behavior); ML is one major path to achieving it (learning from data). Today's deployed systems are overwhelmingly Narrow AI. Within ML, the three core learning paradigms — supervised, unsupervised, and reinforcement learning — differ primarily in what kind of feedback (labels, structure, or reward) is available to the learning algorithm.